Ferran Simon

Curriculum Vitae

Castellbisbal, Spain – ferran@nomis.es – https://ferransimon.github.io/my-cv/

EDUCATION & EMPLOYMENT HISTORY

Staff Full Stack Software Engineer

2017 - Present
Adevinta Spain

Frontend Technical Leadership

- **Frontend Strategic Evolution:** Led the migration from a legacy in-house framework to a modern Open-source stack. This improved server-side scalability, user experience and competitiveness and enabled AI capabilities.
- **Architecture & Monorepos:** Defined the frontend architecture and managed the monorepo strategy by writing and reviewing multiple Architecture Decision Records (ADRs) to standardize patterns across teams.
- **Performance & Reliability:** Created a service reliability plan to fix performance issues like event loop lag, CPU, and memory usage. Used circuit breaker patterns and stress tests to make the application more resilient. Improved Core Web Vitals (CWV) using techniques like code-splitting, lazy loading, transitioning from synchronous react API to the optimized renderToPipeableStream to improve CPU bottlenecks on the server and TTFB metric.
- **Observability & Quality:** Implemented end-to-end observability across services using `traceId` propagation. Set up continuous Core Web Vitals (CWV) monitoring, defined SLOs and actionable monitors using DataDog and created a testing strategy using Vitest and Cypress for unit-tests and end-to-end tests.

Backend, Data & Systems Engineering

- **Monolith Migration:** Led the split of legacy monolithic Oracle databases into a distributed microservices architecture.
- **Event-Driven Architecture & CQRS:** Implemented the CQRS pattern using Kafka as a message broker to decouple services. Built data replication pipelines into read-optimized views or different search engines like OpenSearch to optimize multiple usecases.

- **Distributed Systems:** Solved distributed systems challenges, managing eventual consistency, data integrity, and idempotent replication when processing unsorted event streams.
- **Database Optimization:** Improved database performance for high-volume data by tuning engine configurations and implementing table partitioning. Optimized slow queries by analyzing execution plans, ensuring the use of index scans, and avoiding reading data from disk.

Cross-Functional Impact (CI/CD, AI & Culture)

- **DevOps & CI/CD:** Improved deployment workflows, reducing CI/CD pipeline times from hours to minutes while promoting a "you built it, you run it" culture.
- **AI & Productivity:** Integrated AI tools into the development workflow to improve developer productivity, measuring the results (DX) using DORA metrics.
- **Team Mentorship:** Mentored junior and senior engineers across the full stack, promoting engineering best practices and continuous learning. Introduced the frontend team to AWS and Kubernetes by running multiple technical training sessions.

Software Engineer

2014 - 2017
Blueliv

Developed and maintained a high-performance web crawler using Python and Java, which collected and processed large volumes of data from various sources. Designed and implemented RESTful APIs to expose the collected data to clients, ensuring scalability and reliability. Utilized MongoDB for efficient data storage and retrieval, optimizing database performance for large datasets. Implemented asynchronous task processing using Celery and SQS, improving the efficiency of data processing workflows. Collaborated with cross-functional teams to integrate the crawler with other systems and services, enhancing overall product functionality.

Degree in Telecommunications Engineering

2006 - 2013
Universitat Politècnica de Catalunya

Gained a strong foundation in programming, algorithms, and data structures, as well as knowledge of computer networks and communication systems. Developed my thesis at Northeastern University, Boston, where I published a research paper on a Multi-Path Model and Sensitivity Analysis for Galvanic Coupled Intra-Body Communication Through Layered Tissues.

Degree in Electronics Engineering

2010 - 2013
Universitat Politècnica de Catalunya

Acquired a solid understanding of electronics, circuit design, and signal processing. This degree complemented my telecommunications engineering studies by providing a deeper insight into the hardware aspects of communication systems and electronic devices. Developed my thesis at National Institute of Informatics (NII), Tokyo, where I analyzed GPS data from cars all over Japan to identify accidents caused by a typhoon.

PUBLICATIONS

Multi-Path Model and Sensitivity Analysis for Galvanic Coupled Intra-Body Communication Through Layered Tissues

https://pubmed.ncbi.nlm.nih.gov/25974946/

SKILLS

Programming languages:
JavaScript, TypeScript, Python, Java, Kotlin

Front End:
React, Next.js, CWV, Vitest, Cypress, MVVM

Back End:
Apache Kafka, Distributed Systems, PostgreSQL, Node.js, Spring Boot, Flask, RESTFull API, SQS, Celery, MongoDB, Redis

Craftsmanship:
Clean Architecture, TDD, Tactical DDD, SOLID

DevOps:
AWS, Serverless AWS lambda, CI/CD, Kubernetes, Docker, Monitoring, Debugging

Soft:
Teamwork, Mentoring, Lead by example philosophy, Product Engineering

ABOUT ME

Product-Minded Software Engineer
I am a versatile engineer dedicated to delivering end-to-end features that solve real-world problems. I thrive in the ideation phase, collaborating closely with stakeholders to ensure technical solutions align perfectly with user needs. My approach is defined by a "fail-fast" mentality and a commitment to data-driven decision-making.

I am passionate about engineering excellence and operational peace of mind. I specialize in building robust CI/CD pipelines and utilizing feature flags to ensure seamless roll-outs and instant rollbacks. My goal is always the same: to foster a high-trust environment where the team can deploy with total confidence.